NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE | NTU Academy for Professional and Continuing Education

(SCTP) Advanced Professional Certificate

Data Science and AI

3.6 Time Series Data & Forecasting

Module Overview

3.1 Probability and Statistics for Machine Learning

3.2 Introduction to Machine Learning

3.3 Supervised Learning

3.4 Supervised Learning - Advanced

3.5 Unsupervised Learning

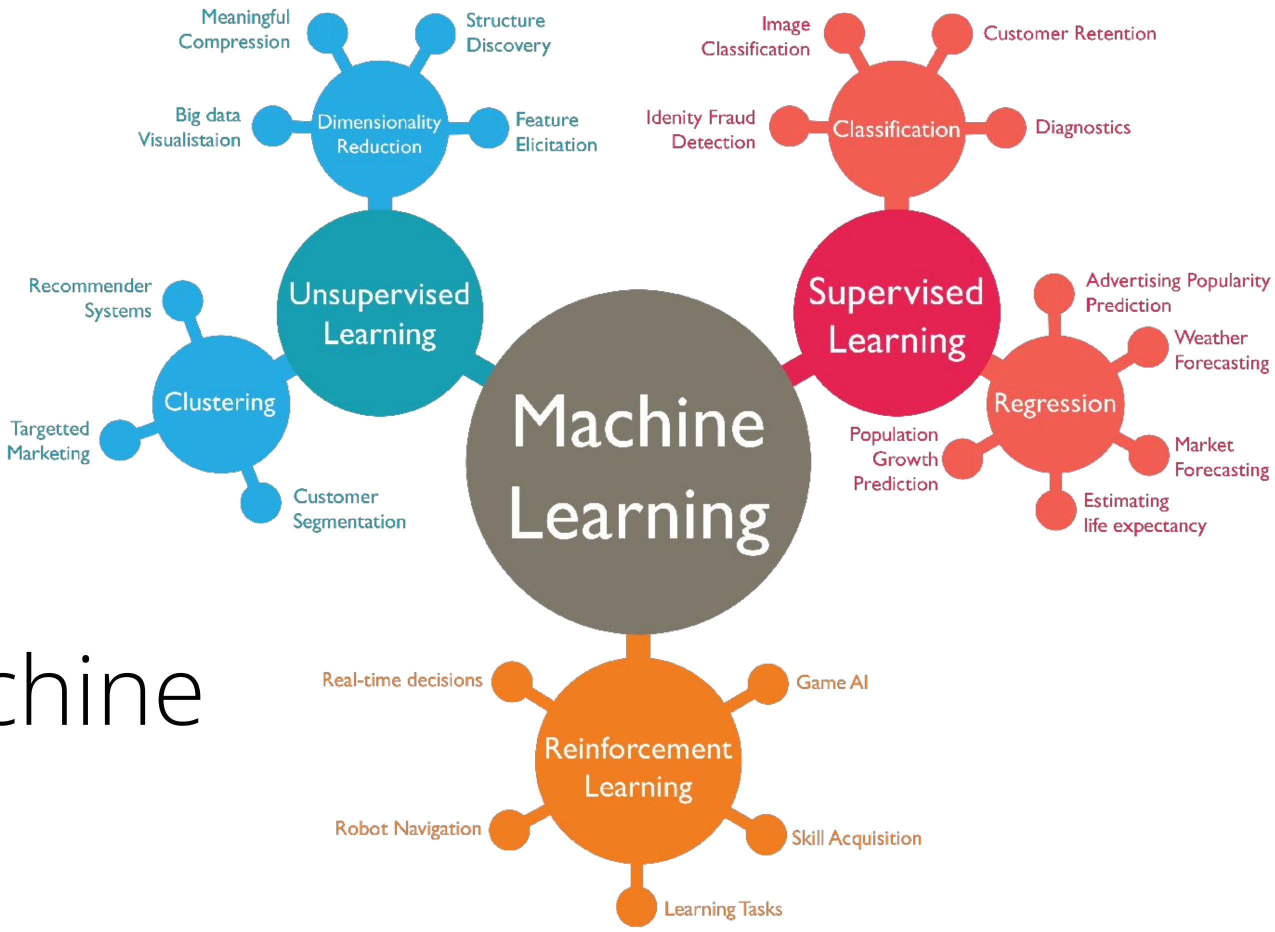
3.6 Time Series Data & Forecasting

3.7 Neural Network & Deep Learning

3.8 Computer Vision (CV)

3.9 Natural Language Processing (NLP)

3.10 NLP - Advanced



Types of Machine Learning

Lesson Objectives

- Plot various time-series charts
- Perform time-series decomposition
- Perform time-series forecasting

Lesson Plan

- **Time Series Data**
- Time Series Visualisation
- Time Series Analysis
- Forecasting Analysis

What is Time Series?

Time series analysis is used when you have data that is collected over time and want to analyze patterns, trends, or make predictions about future values based on historical data. You typically use time series analysis when:

- **Forecasting:** Predicting future values based on past data, such as stock prices, weather patterns, or sales figures.
- **Trend analysis:** Identifying long-term movements or trends in data, such as overall growth or decline in economic indicators, consumer behavior, or company performance.
- **Seasonality:** Understanding and forecasting patterns that repeat at regular intervals, like monthly sales spikes during holidays, daily traffic patterns, or quarterly financial cycles.

What is Time Series?

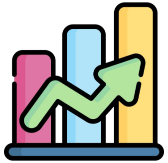
Time series analysis is used when you have data that is collected over time and want to analyze patterns, trends, or make predictions about future values based on historical data. You typically use time series analysis when:

- **Anomaly detection:** Detecting unusual patterns or outliers in data over time, which could indicate fraud, errors, or significant changes in behavior, like a sudden drop in website traffic.
- **Causal impact:** Investigating the effect of external events or interventions on a time series, such as the impact of a marketing campaign on sales or changes in government policy on unemployment.
- **Cycle analysis:** Identifying and modeling recurring cycles that aren't necessarily fixed, like economic booms and recessions, or longer-term cycles in industries.

Time Series Data

- Time series data is data that is recorded over consistent intervals of time.
- Key characteristic: Sequential Nature - Each data point is dependent on the previous points, meaning time order matters.
- Examples:
  - **Financial**: Stock market prices, cryptocurrency trends
  - **Economics**: GDP growth, inflation rates
  - **Business**: Sales revenue, website traffic
  - **Science**: Temperature readings, climate patterns
  - **Healthcare**: Patient vital signs, disease progression

Time Series Data

Pattern Type	Definition	Regularity	Cause	Predictability	Example
Trend 	Long-term increase or decrease in the data	Generally gradual and consistent	Underlying structural changes (e.g., growth, decline)	Moderately predictable	Rising global temperatures over decades
Seasonal 	Repeating short-term pattern at fixed, regular intervals	Regular (e.g., daily, monthly, yearly)	Calendar-based (e.g., weather, holidays)	Highly predictable	Retail spikes in December every year
Cyclic 	Long-term fluctuations that occur at irregular intervals	Irregular	Economic or behavioral forces (not calendar-based)	Less predictable	Business cycles like booms and recessions

Types of Time Series Data

- Time Series Data
- Cross Section Data
- Panel Data

Time Series Data

Time series data consists of **sequential observations** recorded at **regular time intervals** for a single entity. The order of the observations is crucial since past values influence future values.

Examples:

- 📈**Stock Market:** Daily closing prices of a stock.
- 🌡️ **Weather Data:** Hourly temperature readings.
- 🛒**Sales Data:** Monthly sales of a specific product.

Date	Stock Price (USD)
10/3/2025	106.98
11/3/2025	108.86
12/3/2025	115.74
13/3/2025	115.58

Cross Section Data

Cross-sectional data represents observations collected **at a single point in time** across multiple entities. It does not account for temporal changes. Time is NOT a factor.

Examples:

- **Stock Prices:** Daily closing prices of across different stocks.
- **Income Survey:** Household incomes of 1,000 people in 2024.
- **Demographics:** Educational levels of students across different cities.
- **COVID-19 Cases:** Number of infections per country as of June 2024.

Date	NVIDIA	TESLA	AMAZON
10/3/2025	106.98	222.15	194.54

Panel Data

Panel data (or **longitudinal data**) tracks **multiple entities** over **multiple time periods**. It combines aspects of both **time series** and **cross-sectional** data.

Examples:

- **Stock Prices:** Daily closing prices of different stocks in the past 4 days.
- **Economic Growth:** GDP of multiple countries from 2000 to 2024.
- **Employee Salaries:** Annual salaries of employees from different companies from 2015 to 2024.

Date	NVIDIA	TESLA	AMAZON
10/3/2025	106.98	222.15	194.54
11/3/2025	108.86	230.48	196.59
12/3/2025	115.74	248.09	198.89
13/3/2025	115.58	240.68	193.89

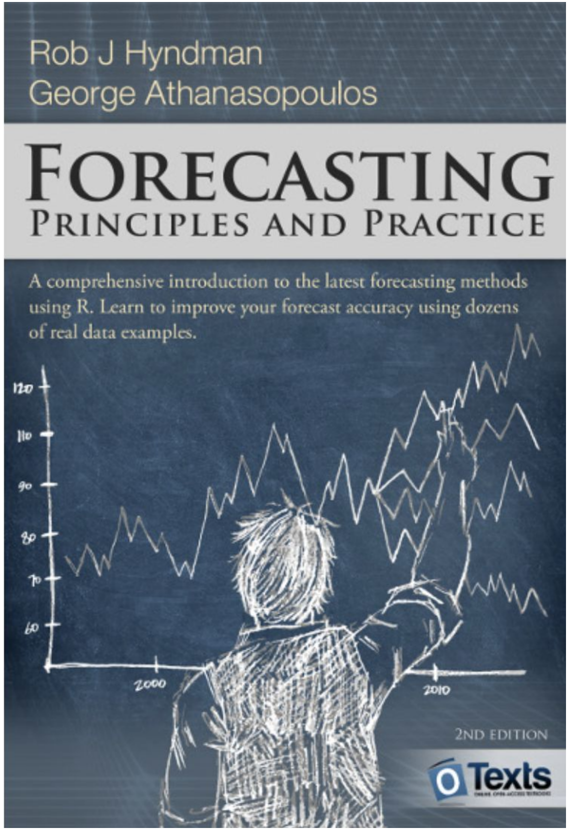
How to Choose the Right Data Type for Your Analysis?

- If you are studying **trends over time** → Use **Time Series Data**.
- If you are comparing **different entities at a single point in time** → Use **Cross-Sectional Data**.
- If you are analyzing **both temporal and cross-sectional patterns** → Use **Panel Data**.

Lesson Plan

- Time Series Data
- **Time Series Visualisation**
- Time Series Analysis
- Forecasting Analysis

Material



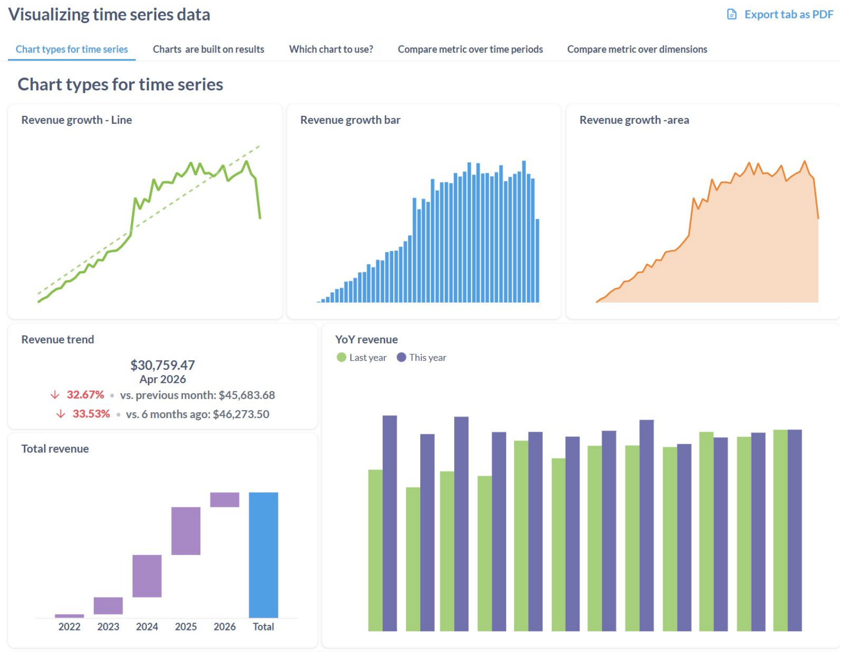
- https://otexts.com/fpp2/
- https://en.wikipedia.org/wiki/Makridakis_Competitions

Time Series Data Visualisation

Visualizing time series data helps to:

- **Identify Trends**: Discover long-term upward or downward movements in data.
- **Understand Seasonality**: Detect periodic patterns like sales spikes during holidays.
- **Spot Anomalies**: Highlight unexpected changes, such as outliers or unusual trends.
- **Communicate Insights**: Share findings effectively with stakeholders.

Example



Lesson Plan

- Time Series Data
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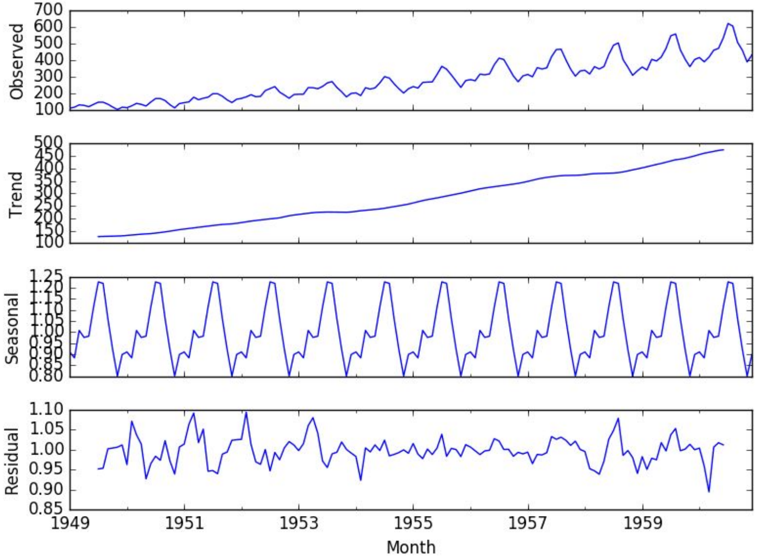
Time Series Decomposition

- Time series decomposition is essentially a process of deconstructing a time series into its constituent parts.
  - **Trend** (long-term direction; increasing / decreasing)
  - **Seasonality** (recurring patterns; monthly, season, etc)
  - **Irregular components** (random noise)
- Understand Patterns: Separates the effects of trend, seasonality, and noise.
- Modeling Readiness: Decomposed data is easier to model and analyze.



https://otexts.com/fpp2/decomposition.html

Time Series Decomposition

2 key methods:

Additive

$$Y = T + S + R$$

- Consistent change on numerical scale
- Used when seasonal variation is independent of the trend.

Multiplicative

$$Y = T * S * R$$

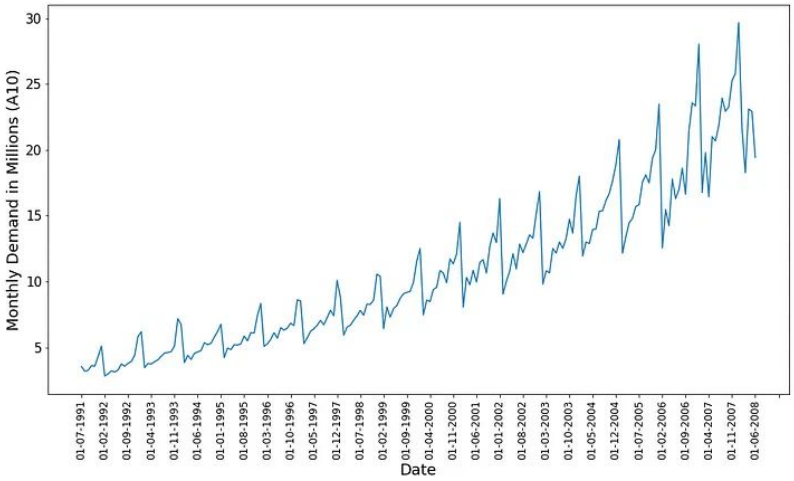
- Change is on a relative scale
- Used when seasonal variation is proportional to the trend.

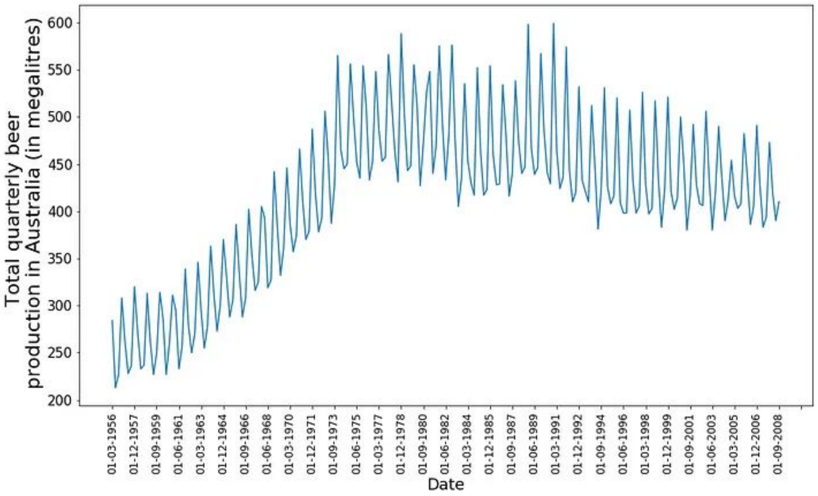
Key Considerations

- ✓ Use multiplicative when seasonal variation grows as the trend increases (peaks and troughs get bigger).
- ✓ Possible to change a multiplicative model to additive via log transformation / Box-Cox Transformation.

Additive? Multiplicative?

Additive? Multiplicative?





Time Series Decomposition

Approach (Classical Approach)

1. Compute Trend Component

using moving / rolling average

2. De-trend the Series

(Y - T) or (Y / T)

3. Compute Seasonal Component

by taking the average of de-trended series for each season

4. Calculate Residual Component

(Y - T – S) or (Y / (T * S))

Other approaches – STL, X11 , SEATS

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Part 1: Visualizing Time-Series Data

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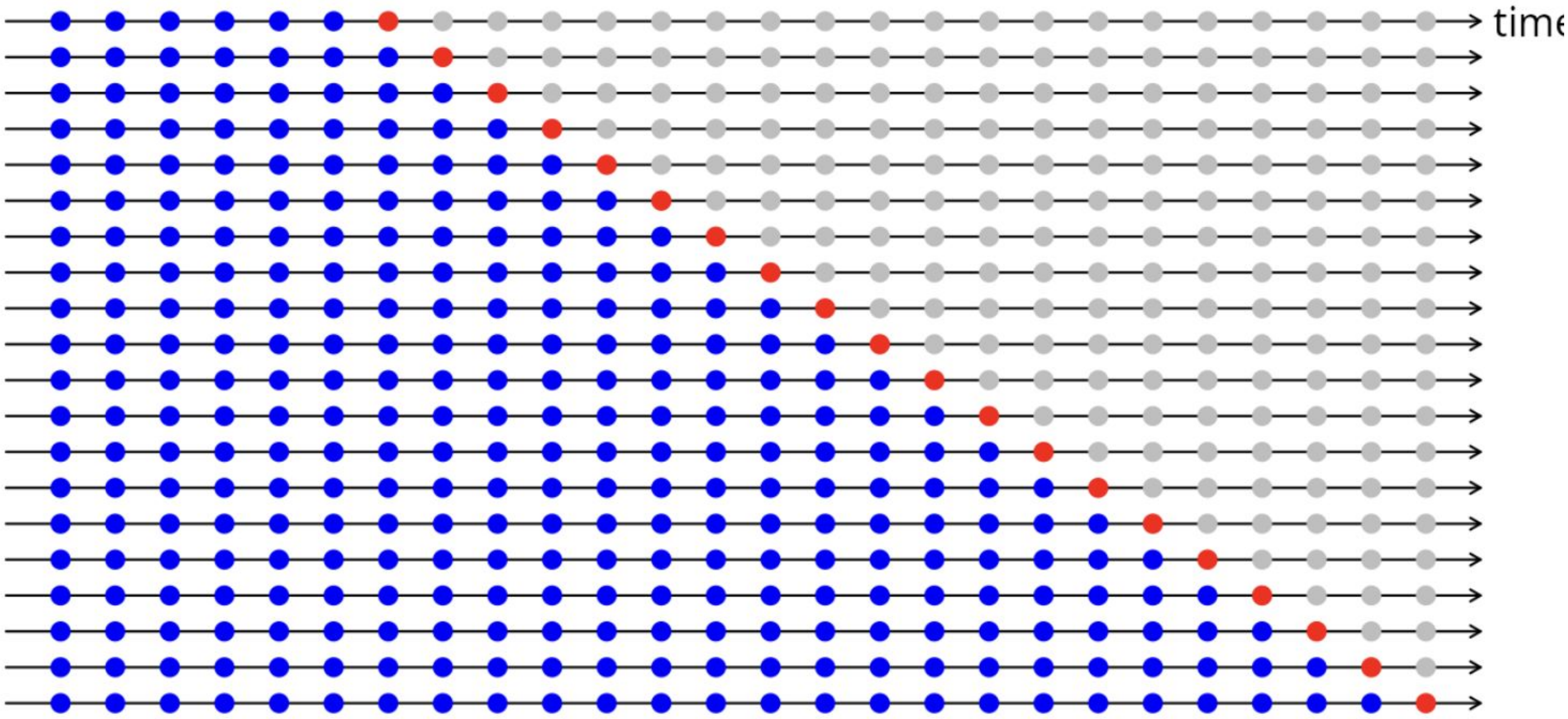
Part 2: Time Series Decomposition

Cross Validation (Chapter 3)



Percentage errors

The percentage error is given by $p_t = 100e_t/y_t$. Percentage errors have the advantage of being unit-free, and so are frequently used to compare forecast performances between data sets. The most commonly used measure is:

$$\text{Mean absolute percentage error: MAPE} = \text{mean}(|p_t|).$$

$$\text{wMAPE} = \frac{\sum_{i=1}^{n} \left(w_i \cdot \frac{|A_i - F_i|}{|A_i|} \right)}{\sum_{i=1}^{n} w_i} = \frac{\sum_{i=1}^{n} \left(|A_i| \cdot \frac{|A_i - F_i|}{|A_i|} \right)}{\sum_{i=1}^{n} |A_i|}$$

Where w_i is the weight, A is a vector of the actual data and F is the forecast or prediction. However, this effectively simplifies to a much simpler formula:

$$\text{wMAPE} = \frac{\sum_{i=1}^{n} |A_i - F_i|}{\sum_{i=1}^{n} |A_i|}$$

$$\mathrm{sMAPE} = \mathrm{mean}\left(200|y_t - \hat{y}_t|/(y_t + \hat{y}_t)\right).$$

Evaluation Metrics

1. **Mean Absolute Error (MAE)**:
   - Explanation: The MAE measures the average magnitude of the errors in a set of forecasts, without considering their direction. It's the average over the test sample of the absolute differences between prediction and actual observation where all individual differences have equal weight.

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

2. **Mean Squared Error (MSE)**:
   - Explanation: The MSE measures the average of the squares of the errors—that is, the average squared difference between the estimated values and the actual value. MSE is more sensitive to outliers than MAE because it squares the errors.

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

3. **Root Mean Squared Error (RMSE)**:
   - Explanation: The RMSE is the square root of the MSE and serves to scale the error to the same units as the forecasted variable. It gives a relatively high weight to large errors.

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

4. **Mean Absolute Percentage Error (MAPE)**:
   - Explanation: The MAPE measures the size of the error in percentage terms. It is calculated as the average of the absolute percentage errors of the forecasts.

$$\text{MAPE} = \frac{100\%}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right|$$

5. **Symmetric Mean Absolute Percentage Error (sMAPE)**:
   - Explanation: The sMAPE is a variation of the MAPE that addresses some of its limitations, such as being undefined when y_i is zero. The sMAPE is symmetric because it equally penalizes positive and negative forecast errors.

$$\text{sMAPE} = \frac{100\%}{n}\sum_{i=1}^{n}\frac{|y_i - \hat{y}_i|}{(|y_i| + |\hat{y}_i|)/2}$$

Code-along

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Part 3: Time-Series Forecasting Evaluation and Metrics

Lesson Plan

- Time Series Data
- Time Series Visualisation
- Time Series Analysis
- **Forecasting Analysis**

Time Series Forecasting Approaches

- AutoRegression (AR)
- Moving Average (MA)
- Autoregressive Integrated Moving Average (ARIMA)

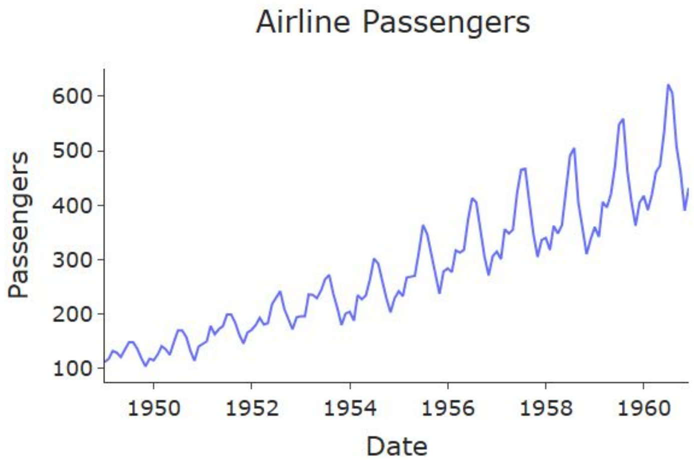
Why Stationarity?

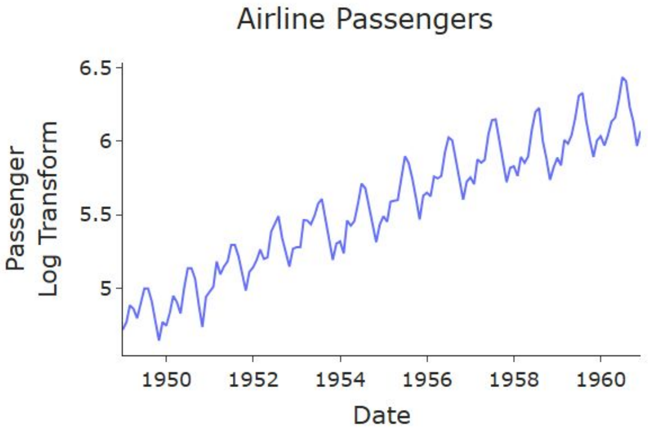
Changing mean and variance over time $\rightarrow$ Changing distribution over time, which makes it harder to model

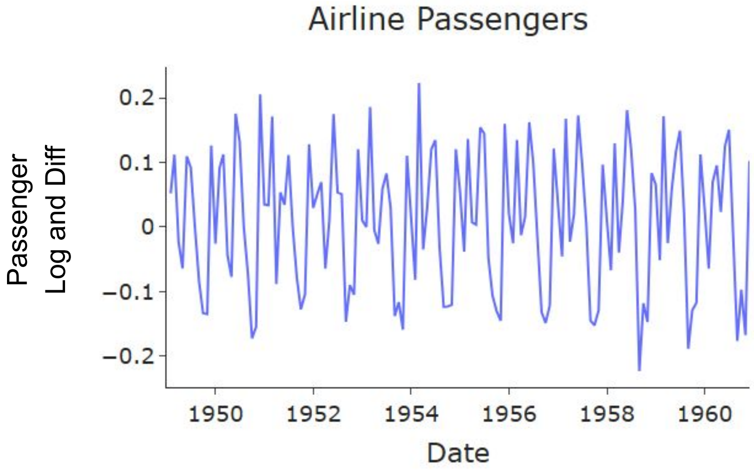
Stationarity means we can assume there is a universal distribution across the data which can be modelled, so it will be easier to forecast and predict.

1. No long term trend
2. No obvious seasonality
3. Constant Mean
4. Constant variance

Stationarity







Autoregressive Model (AR)

An Autoregressive model is a type of time series model where the current value of the series is expressed as a linear combination of its past values, plus some random noise.

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \ldots + \phi_p y_{t-p} + \epsilon_t$$

Where:

- y is the time series forecasted at various time steps,
- ϕ are the fitted coefficients of the lags for the time series,
- ε is the error term (typically normally distributed)
- p is the number of lagged components included in the model, this is also known as the order.

Moving Average (MA)

- Not the same as a “simple moving average” that just averages past values.
- Instead, a Moving Average (MA) model is a time series model where the current value depends on past forecast errors (i.e., the white noise or residuals).

$$y_t = \mu + \epsilon_t + \theta_1\epsilon_{t-1} + \theta_2\epsilon_{t-2} + \ldots + \theta_q\epsilon_{t-q}$$

- Instead of using past values (like AR), the MA model uses past errors, which corrects for temporary noise or irregularities in the data.
- Not to be confused with the rolling mean model, which is also dubbed as a moving average model.

ARIMA Models

Autoregressive Integrated Moving-Average (ARIMA)

ARIMA is a time series forecasting model that combines three components to model patterns in data over time — especially when there's trend or *autocorrelation* (i.e., when past values influence future values).

ARIMA is represented as ARIMA(p, d, q), where:

- p: No. of autoregressive (AR) terms.
- d: No. of differences needed to make the series stationary.
- q: No. of moving average (MA) terms.

ARIMA Formula

The full ARIMA equation is just a linear summation of the three components.

Written in a shorthand way as ARIMA(p, d, q) where p, d and q refer to the order of autoregressive, differencing and moving-averages components respectively.

$$y'_t = c + \phi_1 y'_{t-1} + \ldots + \phi_p y'_{t-p} + \epsilon_t + \theta_1 \epsilon_1 + \ldots + \theta_q \epsilon_{t-q}$$

ARIMA(p, d, q) Parameters

p: No. of autoregressive (AR) terms.

d: No. of differences needed to make the series stationary.

- Determined using repeated KPSS tests.
- Typically try d = 0, 1, 2. Avoid overdamping / noisy models

q: No. of moving average (MA) terms.

Once d is determined, p and q are then chosen by using ACF for MA(q) and PACF for AR(p)

AutoARIMA

Test different combinations of (p, d, q) and chooses the best one using AIC/BIC model selection criteria.

Differencing & Order Selection

- You can perform unit-root test to determine if the data is stationary / non-stationary
    - Determine if differencing is needed
    - Recommend: KPSS Test
    - H0: series is stationary -> check via P-value
- You can select the order empirically by using AIC, BIC, AICC
    - Methods that measures the maximum likelihood estimation (minimize MSE) with some penalty (on the parameters)
    - Ensure differencing (d is set)
- Can be determined automatically via auto arima

Order Selection: Information Criteria

AIC	AICC	BIC
1. -ve log Likelihood + Parameter 2. If large sample is large, almost similar to LOO CV / walk forward CV	Improvement over AIC weakness when sample is small 1. AIC still selects models with large number of parameter if sample is small 2. Converges to AIC if sample is large 3. Bias corrected	1. Penalizes even more on parameters 2. If large sample is large, almost similar to LOO CV / walk forward CV 3. Since penalty is different
Lower better	Lower better	Lower better

AutoARIMA

Hyndman-Khandakar algorithm for automatic ARIMA modelling
1. The number of differences $0 \leq d \leq 2$ is determined using repeated KPSS tests.
2. The values of p and q are then chosen by minimising the AICc after differencing the data d times. Rather than considering every possible combination of p and q, the algorithm uses a stepwise search to traverse the model space.
a. Four initial models are fitted: ◦ $\text{ARIMA}(0, d, 0)$, ◦ $\text{ARIMA}(2, d, 2)$, ◦ $\text{ARIMA}(1, d, 0)$, ◦ $\text{ARIMA}(0, d, 1)$. A constant is included unless $d = 2$. If $d \leq 1$, an additional model is also fitted: ◦ $\text{ARIMA}(0, d, 0)$ without a constant.
b. The best model (with the smallest AICc value) fitted in step (a) is set to be the "current model".
c. Variations on the current model are considered: ◦ vary p and/or q from the current model by ± 1; ◦ include/exclude c from the current model. The best model considered so far (either the current model or one of these variations) becomes the new current model.
d. Repeat Step 2(c) until no lower AICc can be found.

ARIMA Coefficients Interpretation

Coeff. Type	What It Represents
AR (Autoregressive)	The relationship between past values and the current value
MA (Moving Average)	The influence of past forecast errors (residuals) on the current value
Constant (Intercept)	The baseline level of the time series after accounting for trends and seasonality
Differencing (d)	The number of times the data has been differenced to remove trends

ARIMA Advantages

Advantage	Description
📈 Handles Trends	ARIMA can model and forecast time series with trends by differencing the data (d).
🔄 Flexible	Works for a wide variety of time series data (univariate), including those with trend or autocorrelation.
📚 Well-established and Supported	ARIMA is widely used and supported, with extensive documentation and tools like statsmodels and pmdarima.
⏳ Good for Short-Term Forecasting	Ideal for short-to-medium term forecasting, especially when there's trend or autocorrelation.
🧠 Model Interpretability	The model structure is easy to interpret and provides insights into how past values and errors influence future predictions.
🛠️ Widely Supported in Software	Tools like Python's statsmodels and R make it easy to implement and tune ARIMA models.

ARIMA Disadvantages

Limitation	Description
🚫 Requires Stationarity	ARIMA requires the data to be stationary, meaning the mean, variance, and autocorrelation should not change over time. Preprocessing is needed to achieve this (e.g., differencing).
⚠️ Limited to Linear Relationships	Assumes future values are linear combinations of past values — it can't capture nonlinear patterns.
🏃 Short Memory	ARIMA models only look back a fixed number of lags (p) — it may not capture long-term dependencies unless the order of p is large.
❓ Model Order Selection Can Be Tricky	Choosing the right parameters (p, d, q) can be subjective and require model selection methods like AIC/BIC.
🌪️ Poor Performance with Structural Breaks or Shocks	Struggles when there are sudden changes, structural breaks, or outliers (e.g., financial crises or natural disasters).
🔊 Sensitive to Noise and Outliers	ARIMA models can be heavily influenced by outliers, particularly in small datasets.

Key Takeaway

- Time series forecast is typically a separate field
- M4 showed that hybrid with ML + traditional methods perform well and won the competition
- Time series can be done using traditional ML but needs very good feature engineering
    - Complex modern models struggles at trends
    - Need to model autoregressive features
- Explore more:
    - ETS: holt winter’s triple exponential smoothing
    - Prophet (general additive model + bayesian statistics)

Code-along

Jupyter Notebook

Part 4: Forecasting Models (ARIMA)

\+ Class Activity